

IT3091 — Machine Learning: Assignment Data Dictionary

Bank Marketing Dataset (bank-full.csv)

Full variable list

#	Variable	Type	Role	Description	Values / range	Data quality note
1	age	Integer	Feature	Age of the client	18 – 95 (approx.)	No missing values
2	job	Categorical	Feature	Type of job	admin., blue-collar, entrepreneur, housemaid, management, retired, self-employed, services, student, technician, unemployed, unknown (12 categories)	'unknown' is a real category — decide keep vs impute vs drop
3	marital	Categorical	Feature	Marital status	divorced, married, single (divorced = divorced or widowed)	No missing / unknown values
4	education	Categorical	Feature	Education level	primary, secondary, tertiary, unknown	'unknown' present — same treatment decision as job
5	default	Binary	Feature	Has credit in default?	yes / no	Heavily imbalanced toward 'no'
6	balance	Integer	Feature	Average yearly account balance	Euros; can be negative (overdrawn)	Right-skewed with outliers — consider capping / log transform
7	housing	Binary	Feature	Has a housing loan?	yes / no	No missing values
8	loan	Binary	Feature	Has a personal loan?	yes / no	No missing values

#	Variable	Type	Role	Description	Values / range	Data quality note
9	contact	Categorical	Feature	Contact communication type	cellular, telephone, unknown	'unknown' likely means contact type not recorded — treat as its own category
10	day	Integer	Feature	Last contact day of the month	1 – 31	No year given, so cannot be combined into a full date
11	month	Categorical	Feature	Last contact month of year	jan – dec (3-letter codes)	Combine with 'day' to sequence contacts within a year
12	duration	Integer	Excluded	Last contact duration, in seconds	0 and up	EXCLUDE from model — only known after the call happens; including it causes target leakage (per Initial Submission)
13	campaign	Integer	Feature	Number of contacts performed during this campaign for this client	1 and up (includes the last contact)	Right-skewed — most clients contacted only 1–3 times
14	pdays	Integer	Feature	Days since the client was last contacted from a previous campaign	-1 = never previously contacted; otherwise 0+	-1 is a sentinel, not a true numeric value — engineer a 'was_previously_contacted' flag before using pdays numerically
15	previous	Integer	Feature	Number of contacts performed before this campaign, for this client	0 and up	Mostly 0 (majority never contacted before); right-skewed

#	Variable	Type	Role	Description	Values / range	Data quality note
16	poutcome	Categorical	Feature	Outcome of the previous marketing campaign	failure, other, success, unknown	'unknown' dominates (most clients have no prior campaign) — expected, not a data error
17	y	Binary	Target	Has the client subscribed to a term deposit?	yes / no	Target variable. Imbalanced: ~11–12% 'yes' — plan metrics/resampling accordingly

Unit of analysis

One row = one client, represented by their most recent campaign contact record. There is no client ID column, so rows cannot be linked across multiple campaigns.